

AlphaVerify: Mathematical Specification

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1 Purpose and scope

AlphaVerify asks a narrow question about a market condition: *when the condition holds, does price become more or less likely to reach a given distance up or down within a given time, and is that change larger than chance would produce?* It answers in five stages:

Stage	Question	Result
1 Measure	How often was each barrier touched, over all dates and within each feature bin?	B, Y_k
2 Compare	How much does a bin change those chances?	$Z_k = Y_k - B$
3 Validate	Could a market without memory produce as large a change?	$\hat{p}$
4 Select	Which bins pass, and how many would pass by chance?	selected bins, q
5 Combine	What do the passing conditions that hold today imply together?	P, J

This document explains what each stage computes and why it is built that way. Formulas are written at the level needed to follow the idea; the exact numerical conventions are collected briefly in Section 9.

What the comparison establishes

Validation compares each result with a fitted simulator in which prices have no memory.

What a small p-value means

This simulator rarely produces a score as large as the observed one.

It does not by itself show out-of-sample performance, an edge after costs, or a calibrated 5% false-positive rate in real markets, which do have memory (Section 6).

2 Central concepts

2.1 The Surface as the computational unit

Everything in AlphaVerify is laid out on two ordered grids: a barrier grid Δ and a horizon grid Θ .

- A *barrier* $\delta \in \Delta$ is a distance from the starting date's close. For example, $\delta = -0.05$ is 5% below that close, and $\delta = +0.05$ is 5% above it.
- A *horizon* $\theta \in \Theta$ is a number of bars ahead. For example, $\theta = 10$ asks whether the barrier is touched at any point within the next ten bars, not only at the end of the tenth.

A Surface assigns a real number to every pair of a barrier and a horizon. All Surfaces therefore live in one space:

The Surface space

$$\mathcal{S} = \mathbb{R}^{\Delta \times \Theta}, \quad F \in \mathcal{S}, \quad F(\delta, \theta) \in \mathbb{R}.$$

$\mathcal{S}$ is the space of Surfaces, and F is one Surface. As an array it has $|\Delta|$ rows and $|\Theta|$ columns, but mathematically the whole grid is a single element of $\mathcal{S}$: one point in this space. That element is the unit the pipeline operates on. It is measured, compared, scored, and selected as one object, even though it is built cell by cell.

Blank cells

A cell whose value would rest on fewer than 30 dates is left blank. A blank cell has no value; it is not zero. A Surface keeps its full shape even when some of its cells are blank. Formally, a partially supported Surface is an element $F \in \mathcal{S}$ together with its set of supported cells $\mathcal{M} \subseteq \Delta \times \Theta$; values of F outside $\mathcal{M}$ are never used.

2.2 Observed and synthetic histories

Everything is computed on a price history: a sequence of bars with open, high, low, close, and volume. The observed history is the real market. Stage 3 also builds $I = 1,000$ synthetic histories (Section 6.3) and runs each of them through exactly the same measurement and scoring. Index the histories by $i = 0, 1, \dots, I$, with $i = 0$ the observed one. Because the rules are identical for every i , the formulas in this document describe one history and leave i out; it appears only where observed and synthetic results are compared (Stage 3).

2.3 The objects carried through the pipeline

Four Surface objects, one common space

$$X_t, \quad Y_k, \quad B, \quad Z_k \in \mathcal{S}.$$

Their subscripts say which member of $\mathcal{S}$ is meant, a date t or a bin k ; their coordinates are always (δ, θ) . Their values mean different things, but they are elements of the same space. On supported cells:

Outcome Surface

$X_t(\delta, \theta) \in \{0, 1\}$ is 1 if barrier δ was touched within θ bars after date t , and 0 if not.

Conditional probability Surface

$Y_k(\delta, \theta) \in [0, 1]$ is the share of dates in feature bin k that were followed by that touch. For example, 0.60 means 60% of them.

Baseline Surface

$B(\delta, \theta) \in [0, 1]$ is the same share over all dates, whatever the feature says. There is one baseline per history, shared by every bin.

Shift Surface

$Z_k(\delta, \theta) = Y_k(\delta, \theta) - B(\delta, \theta) \in [-1, 1]$ is how much the bin changes the chance of the touch. 0.60 against a baseline of 0.40 gives +0.20, shown as +20%: a 20-percentage-point difference, not a relative change.

Y_k and B are the same calculation over different groups of dates: the dates in bin k , or all dates. The baseline is not one of the bins.

Object	One Surface per	Meaning of a cell (δ, θ)
X_t	Date	Whether the barrier was touched
Y_k	Bin	Touch rate within the bin
B	History	Touch rate over all dates
Z_k	Bin	Bin minus baseline

2.4 The score

Stage 3 condenses each shift Surface into two scalars:

One score and one support flag per bin

$$Q_k \in \mathbb{R}_{\geq 0}, \quad V_k \in \{0, 1\}.$$

Score

Q_k measures how lopsided the shift is between barriers above and below the close. It is not a probability and can exceed one.

Support flag

$V_k = 1$ when at least one cell of the bin can be scored. An unsupported bin is stored as $Q_k = 0$, $V_k = 0$ and is not tested; a supported bin can also score $Q_k = 0$, with $V_k = 1$.

The central computation

$$\{X_t\}_t \subset \mathcal{S} \xrightarrow{\text{touch rates}} Y_k, \quad B \in \mathcal{S} \xrightarrow{\text{subtract baseline}} Z_k \in \mathcal{S} \xrightarrow{\text{score}} (Q_k, V_k) \in \mathbb{R}_{\geq 0} \times \{0, 1\}.$$

Every step until scoring maps Surfaces to Surfaces. Scoring is the one step that leaves $\mathcal{S}$: it turns a whole Surface into a single number and its support flag. A bin's score also depends on the other bins of the same feature: a cell counts only where at least two bins have enough dates (Section 6.1).

3 Notation

The same few symbols appear throughout.

Symbol	Meaning
$i \in \{0, \dots, I\}$	A history: 0 is observed, $1, \dots, I$ are synthetic ($I = 1,000$).
$t \in \mathcal{T}$	A date (one bar); $t + j$ is j bars later.
x_t	The feature's value on date t .
k	A feature bin, counted from 0 (lowest values) upward.
$\delta \in \Delta, \theta \in \Theta$	A barrier, as a fraction of the close, and a horizon, in bars, from their ordered grids.
$\mathcal{S} = \mathbb{R}^{\Delta \times \Theta}$	The Surface space; every Surface is one element of it.
$X_t, Y_k, B, Z_k \in \mathcal{S}$	Outcome, conditional, baseline, and shift Surfaces.
$\mathcal{E}_\theta, A_k$	Dates with θ bars of future data; dates whose feature value falls in bin k .
$\Pi[A] \in \mathcal{S}$	The touch-rate Surface of a group of dates A .
$N_k(\theta), N(\theta)$	$ A_k \cap \mathcal{E}_\theta $ and $ \mathcal{E}_\theta $: the dates behind a cell of Y_k and of B .
$\Delta_\pm, \mathcal{U}_k$	Barriers with a unique mirror; the usable cells of bin k (Stage 3).
$Q_k, V_k, \hat{p}$	A bin's score, its support flag, and its Monte Carlo p-value (Stage 3).
m, q	The number of tested bins and a bin's q-value (Stage 4).
$\tilde{p}, P, J$	A smoothed rate, the naive Bayes Surface, and the historical joint Surface (Stage 5).

The support threshold

$$n_{\min} = 30$$

A cell measured from fewer than $n_{\min}$ dates is blank: it has no value, not a value of zero.

Counts shrink at longer horizons, because the last θ dates of a history have no complete window ahead of them. Long horizons therefore go blank first.

4 Stage 1: measurement of probability surfaces

Stage 1 turns prices into touch rates. It first records, for every date, whether each barrier was reached, then groups the dates by the feature and averages.

Stage 1 at a glance

$$\text{prices} \xrightarrow{\text{was it touched?}} X_t \text{ for every date } \xrightarrow{\text{average by bin, and over all dates}} (Y_k, B).$$

4.1 From prices to an outcome Surface

Start from the close on date t . Barrier δ sits at the price $\text{Close}_t(1 + \delta)$. It is touched within θ bars when some later low reaches it (a barrier below) or some later high reaches it (a barrier above). “Later” means bars $t + 1$ to $t + \theta$; bar t itself never counts.

The outcome Surface: was the barrier touched?

$$X_t(\delta, \theta) = \begin{cases} \mathbf{1} \left[\min_{1 \leq j \leq \theta} \text{Low}_{t+j} \leq \text{Close}_t(1 + \delta) \right], & \delta < 0, \\ \mathbf{1} \left[\max_{1 \leq j \leq \theta} \text{High}_{t+j} \geq \text{Close}_t(1 + \delta) \right], & \delta \geq 0, \end{cases} \quad t \in \mathcal{E}_\theta,$$

where $\mathbf{1}[\cdot]$ is 1 when its statement is true and 0 otherwise.

Highs and lows are used, not closes, so a barrier counts as touched even if price comes back before the bar ends. The zero barrier asks whether any later high reached the starting close. Only dates with a complete window count:

$$\mathcal{E}_\theta = \{t \in \mathcal{T} : \text{bar } t + \theta \text{ exists}\},$$

so the last θ dates of a history are left out at horizon θ .

X_t depends only on prices. The feature never changes an outcome; it only decides which dates are averaged together.

4.2 From feature values to bins

Bins are quantiles of the feature. With the default of $K = 10$ bins, each bin holds about a tenth of the dates that have a feature value, from the lowest values to the highest. Dates without a feature value, such as during an indicator’s warm-up period, belong to no bin.

Precisely, the edges are the interior quantiles of the finite feature values,

$$e_j = \text{the } \frac{j}{K}\text{-quantile of } \{x_t\}, \quad j = 1, \dots, K - 1,$$

with repeated edges removed and only edges in $(\min x, \max x]$ kept. A date’s bin is the number of edges strictly below its value, and a bin is the set of dates assigned to it:

Bin assignment

$$k(t) = \#\{j : e_j < x_t\}, \quad A_k = \{t : x_t \text{ is finite and } k(t) = k\}.$$

A value exactly on a bin edge is not strictly above it, so it goes to the lower bin. With edges 2 and 5:

x	2	3	5	6
bin	0	1	1	2

A heavily tied feature can end up with fewer bins than requested, because repeated values produce repeated edges, which merge. Asking for four bins from 60 zeros and 20 ones gives the quantiles $(0, 0, 0.25)$; only 0.25 survives, so there are two bins.

4.3 Averaging outcome surfaces

A touch rate is the share of a group of dates that were followed by a touch. For any group of dates A , it defines a whole Surface:

$$\Pi[A](\delta, \theta) = \frac{1}{|A \cap \mathcal{E}_\theta|} \sum_{t \in A \cap \mathcal{E}_\theta} X_t(\delta, \theta), \quad \text{blank when } |A \cap \mathcal{E}_\theta| < n_{\min}.$$

Π maps a group of dates to an element of $\mathcal{S}$. Stage 1 applies it to two kinds of group:

One touch rate, two groups of dates

$$Y_k = \Pi[A_k], \quad B = \Pi[\mathcal{T}].$$

In words: the share of dates in bin k , or of all dates, that were followed by a touch. The dates behind a cell are $N_k(\theta) = |A_k \cap \mathcal{E}_\theta|$ and $N(\theta) = |\mathcal{E}_\theta|$.

Both count only dates with θ bars of future data, and both are blank below $n_{\min} = 30$ dates.

The baseline includes every such date, even those without a feature value. The baseline is therefore not simply the average of the bins. Each synthetic history has its own baseline.

4.4 Worked example: measuring a Surface

Take five barriers and seven daily horizons:

$$\Delta = \{-2\%, -1\%, 0\%, +1\%, +2\%\}, \quad \Theta = \{1, 2, 3, 4, 5, 6, 7\}.$$

Assume a market with a bar every calendar day, so seven bars cover one week.

Start from a close of 100. The five barrier prices are then 98, 99, 100, 101, 102. Suppose later prices first reach 100 on day 1, 101 on day 2, 99 on day 3, 102 on day 5, and 98 on day 6. This gives the following outcome Surface:

$\delta \backslash \theta$	1	2	3	4	5	6	7
-2%	0	0	0	0	0	1	1
-1%	0	0	1	1	1	1	1
0%	1	1	1	1	1	1	1
$+1\%$	0	1	1	1	1	1	1
$+2\%$	0	0	0	0	1	1	1

For example, $X_t(+2\%, 4) = 0$ but $X_t(+2\%, 5) = 1$: the price had not reached 102 within four days, but had within five. Once a row becomes 1, it stays 1 at longer horizons, even if price later reverses.

Now collect 100 starting dates in the same bin k , each with seven days of future data. Each date contributes one such 5×7 outcome Surface. Averaging them cell by cell gives

$\delta \backslash \theta$	1	2	3	4	5	6	7
-2%	0.05	0.10	0.15	0.20	0.25	0.30	0.35
-1%	0.15	0.20	0.25	0.30	0.35	0.40	0.45
0%	0.70	0.75	0.80	0.85	0.90	0.95	1.00
$+1\%$	0.25	0.30	0.35	0.40	0.45	0.50	0.55
$+2\%$	0.15	0.20	0.25	0.30	0.35	0.40	0.45

The cell $Y_k(+2\%, 5) = 0.35$ means that 35 of the 100 dates were followed by a touch of the $+2\%$ barrier within five days. In real histories, longer horizons usually rest on slightly fewer dates.

5 Stage 2: comparison with the baseline

Stage 2 asks how much a bin changes the chance of each touch.

5.1 Baseline subtraction

The shift Surface

$$Z_k(\delta, \theta) = Y_k(\delta, \theta) - B(\delta, \theta),$$

cell by cell. A cell blank in either Surface is blank in Z_k .

Positive cell (red)

The barrier was touched more often in the bin than over all dates.

Negative cell (blue)

The barrier was touched less often.

Zero cell (white)

The bin made no difference at that barrier and horizon.

Every synthetic history subtracts its own baseline, never the observed one.

5.2 Worked example: subtracting the baseline

Continue with the conditional Surface Y_k of Section 4.4. Suppose averaging over all dates of the same history gives the baseline

$\delta \backslash \theta$	1	2	3	4	5	6	7
-2%	0.15	0.20	0.25	0.30	0.35	0.40	0.45
-1%	0.25	0.30	0.35	0.40	0.45	0.50	0.55
0%	0.70	0.75	0.80	0.85	0.90	0.95	1.00
$+1\%$	0.15	0.20	0.25	0.30	0.35	0.40	0.45
$+2\%$	0.05	0.10	0.15	0.20	0.25	0.30	0.35

Subtracting cell by cell gives one complete shift Surface:

$\delta \backslash \theta$	1	2	3	4	5	6	7
-2%	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10
-1%	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10
0%	0	0	0	0	0	0	0
$+1\%$	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10
$+2\%$	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10

For example, at $(+2\%, 5)$ the bin's rate is 35% and the baseline is 25%, so $Z_k(+2\%, 5) = 0.35 - 0.25 = +0.10$. At $(-2\%, 5)$ it is $0.25 - 0.35 = -0.10$. The zero-barrier rates are equal, so their shifts are zero. In this bin the price reached upside barriers more often and downside barriers less often, at every horizon.

6 Stage 3: validation by null comparison

Stage 2 produced a shift Surface for every bin. Stage 3 asks one question of each: *could a market in which the feature cannot matter have produced a Surface this lopsided?* It condenses each Surface into one score, builds $I = 1,000$ synthetic histories that have no memory, scores the same bin on each, and counts how often a synthetic history scores at least as high.

After the score is defined (Section 6.1), fix one feature and one bin and drop the bin index. Write Z for the bin's observed shift Surface, Q_{obs} and V_{obs} for its observed score and support flag, and Q_i and V_i for the same quantities on synthetic history $i = 1, \dots, I$.

Stage 3 at a glance

$$\begin{array}{lcl}
\text{observed history} & \xrightarrow{\text{Stages 1-2}} Z & \xrightarrow{\text{score}} Q_{\text{obs}} \\
\text{synthetic histories } 1, \dots, I & \xrightarrow{\text{Stages 1-2}} Z_1, \dots, Z_I & \xrightarrow{\text{score}} Q_1, \dots, Q_I \\
\text{compare} & \xrightarrow{\text{how often } Q_i \geq Q_{\text{obs}}} & \hat{p}
\end{array}$$

6.1 Scoring a shift Surface

The score measures how differently a condition moves the chance of an up-move and a down-move of the same size. Each barrier is compared with its mirror image at the same horizon, $+\delta$ with $-\delta$.

In this subsection the bin index k stays, because which cells count depends on the other bins of the same feature.

Only barriers with a mirror can be compared:

$$\Delta_{\pm} = \{\delta \in \Delta : \delta \neq 0, \text{ both } \delta \text{ and } -\delta \text{ appear exactly once in } \Delta\}, \quad \delta_{\max} = \max_{\delta \in \Delta_{\pm}} |\delta|.$$

The score: asymmetry between mirror barriers

$$Q_k = \sum_{(\delta, \theta) \in \mathcal{U}_k} \frac{|\delta|}{2\delta_{\max}} |Z_k(\delta, \theta) - Z_k(-\delta, \theta)|, \quad V_k = \mathbf{1}[\mathcal{U}_k \neq \emptyset],$$

where $\mathcal{U}_k \subseteq \Delta_{\pm} \times \Theta$ is the set of usable cells of bin k , defined below.

Read the formula one piece at a time:

Mirror difference

$|Z_k(\delta, \theta) - Z_k(-\delta, \theta)|$ is large when the condition makes one side more likely to be touched and the other side less likely. If both sides shift by the same amount, it is zero. The absolute value ignores which side wins.

Distance weight

$|\delta|/\delta_{\max}$ makes far barriers count more, in proportion to their distance: a $\pm 2\%$ pair counts twice as much as a $\pm 1\%$ pair.

The factor 1/2

Every pair appears twice in the sum, once at $+\delta$ and once at $-\delta$. Halving counts each pair once.

Which cells count

A cell is *usable* for bin k when three things hold at that barrier and horizon: bin k has at least $n_{\min} = 30$ dates, both mirror shifts exist, and at least one other bin of the same feature meets the same two conditions. First collect the bins that are measurable at a mirror pair,

$$\mathcal{L}(\delta, \theta) = \{j : N_j(\theta) \geq n_{\min}, Z_j(\delta, \theta) \text{ and } Z_j(-\delta, \theta) \text{ are not blank}\}, \quad \delta \in \Delta_{\pm},$$

then keep the cells where bin k is one of at least two such bins:

Usable cells

$$\mathcal{U}_k = \{(\delta, \theta) \in \Delta_{\pm} \times \Theta : k \in \mathcal{L}(\delta, \theta), |\mathcal{L}(\delta, \theta)| \geq 2\}.$$

A cell therefore counts only where the feature splits the dates into at least two measurable groups. The zero barrier is not in $\Delta_{\pm}$ and never counts.

A bin is *supported* ($V_k = 1$) when at least one of its cells is usable. An unsupported bin is stored with score 0, but “unsupported” and “scored zero” are different facts: a supported bin whose mirror shifts are equal also scores 0. Losing some horizons does not make a bin unsupported; only losing every usable cell does.

The grid should be symmetric. A barrier outside $\Delta_{\pm}$ is ignored, and it does not affect $\delta_{\max}$: adding an unpaired +5% row to a $\pm 2\%$ grid leaves every weight unchanged. A grid with $\Delta_{\pm} = \emptyset$ gives every bin score 0 and no support. That outcome is a configuration mistake, not evidence of no effect.

What the score does not do

The score is a weighted sum of probability differences, not a probability, and it can exceed one. It is not divided by the number of usable cells, so a bin with more usable cells can reach a larger score. Every usable cell counts equally whether it rests on 31 dates or 3,000. For scale, a rate measured from 30 independent dates has a standard error of about 0.09 near 50%, and overlapping forward windows make the real uncertainty larger. The score does not weight cells by this precision.

6.2 Worked example: scoring the shift

For the running example, take the Stage 2 Surface as the observed Z . Its two negative-barrier rows are -0.10 and its two positive-barrier rows are $+0.10$ at every horizon. The bin has 100 dates at every horizon, and at least one other bin is measurable everywhere, so every nonzero cell is usable.

Each row is paired with its mirror and weighted by its distance:

Barrier	-2%	-1%	0%	$+1\%$	$+2\%$
Mirror	$+2\%$	$+1\%$	none	-1%	-2%
Weight $ \delta /\delta_{\max}$	1	0.5	0	0.5	1

Every mirror difference is $|-0.10 - (+0.10)| = 0.20$. A 1% cell contributes $\frac{1}{2} \cdot 0.5 \cdot 0.20 = 0.05$, and a 2% cell contributes $\frac{1}{2} \cdot 1 \cdot 0.20 = 0.10$. Laid out on the grid:

$\delta \backslash \theta$	1	2	3	4	5	6	7
-2%	0.10	0.10	0.10	0.10	0.10	0.10	0.10
-1%	0.05	0.05	0.05	0.05	0.05	0.05	0.05
0%	0	0	0	0	0	0	0
$+1\%$	0.05	0.05	0.05	0.05	0.05	0.05	0.05
$+2\%$	0.10	0.10	0.10	0.10	0.10	0.10	0.10

Each horizon contributes $0.10 + 0.05 + 0 + 0.05 + 0.10 = 0.30$, and seven horizons give

$$Q_{\text{obs}} = 7 \times 0.30 = 2.10.$$

Stage 3 compares this one number with the synthetic scores; it does not test the 35 cells separately.

Worked boundary cases: support is not a score

Use one horizon and the barriers $\pm 2\%$, so both weights are 1. Start from two measurable bins, with shifts -0.10 and $+0.10$ in this bin, and change one thing at a time:

Case	What changes	Score	Supported
Reference	Nothing: two contributions of 0.10.	0.20	yes
Too few dates	This bin has 29 dates, so its shifts do not exist. The other bin, now the only measurable one, loses support too.	0	no
Equal shifts	Both shifts are +0.10: no asymmetry, but fully measured.	0	yes
No mirror	The barriers are -2% and $+1\%$, so nothing pairs up.	0	no
Heavy ties	Four bins requested from 60 zeros and 20 ones collapse into two bins of 60 and 20 dates. The 20-date bin is too small, which leaves the other bin alone.	0	no

6.3 The synthetic-history model

A synthetic history should look like the market one bar at a time, but have no memory: nothing in past bars can say anything about the next one. Then no feature, whether computed from past bars or taken from outside data, can predict the next bar, so any score it earns is chance.

Each bar is described by four log components:

Component	Definition	Part of the bar
g_t^{open}	$\log\left(\frac{\text{Open}_t}{\text{Close}_{t-1}}\right)$	Gap from the previous close
g_t^{close}	$\log\left(\frac{\text{Close}_t}{\text{Open}_t}\right)$	Body, open to close
g_t^{high}	$\log\left(\frac{\text{High}_t}{\max(\text{Open}_t, \text{Close}_t)}\right)$	Upper wick
g_t^{low}	$\log\left(\frac{\text{Low}_t}{\min(\text{Open}_t, \text{Close}_t)}\right)$	Lower wick

Stack them into one vector per bar, $g_t = (g_t^{\text{open}}, g_t^{\text{close}}, g_t^{\text{high}}, g_t^{\text{low}}) \in \mathbb{R}^4$. Let μ and Σ be the sample mean and covariance of the observed vectors, then draw every synthetic bar on its own:

One synthetic bar: realistic shape, no memory

$$g_1, \dots, g_{|\mathcal{T}|} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \Sigma).$$

Within a bar the four components keep their observed correlations; across bars they are independent. Chain the bars into prices, starting from the first observed close as Close_0 :

$$\begin{aligned} \text{Open}_t &= \text{Close}_{t-1} e^{g_t^{\text{open}}}, & \text{Close}_t &= \text{Open}_t e^{g_t^{\text{close}}}, \\ \text{High}_t &= \max(\text{Open}_t, \text{Close}_t) e^{g_t^{\text{high}}}, & \text{Low}_t &= \min(\text{Open}_t, \text{Close}_t) e^{g_t^{\text{low}}}. \end{aligned}$$

Each synthetic history has as many bars as the observed one, and volume is copied from the observed history. A fixed seed makes the ensemble reproducible.

What the null keeps

The average size of each part of a bar, and how the four parts move together within one bar.

What the null discards

Every link between bars: autocorrelation, volatility clustering, and regimes. It also has no liquidity, execution, or trading costs.

What this null can and cannot show

Real markets have memory; this null has none. A feature that tracks persistent volatility or a regime can therefore beat the null without predicting anything beyond “volatile periods stay volatile.” Clearing means *beats a memoryless market*, not *carries a usable edge*.

The normal draw can also produce impossible bars. An upper wick of -1% puts the high below the body; the reconstruction does not repair such bars.

A null with memory, such as a block bootstrap, would fix the first problem but could also keep the very feature–future relationship under test. Any replacement has to say what it removes, what it keeps, and how its calibration is checked.

6.4 Repeating measurement on each null path

Features follow one of two policies on synthetic histories:

OHLCV-derived feature (recomputed)

Recompute the feature and its quantile edges on each synthetic history.

External, calendar, or workspace-extension feature (fixed)

These cannot be rebuilt from synthetic prices, so every synthetic history reuses the observed values and cutoffs.

Each synthetic history then goes through Stages 1–2 and the scoring above exactly as the observed history did, giving $Q_1, \dots, Q_I$. “The same bin” means the same position, such as the lowest decile, on every history. For a recomputed feature, its cutoffs are that history’s own deciles, so they differ between histories. Volume is copied from the observed history even when a feature uses it, so its link to synthetic prices is lost.

6.5 The test and decision rule

The p-value is the observed score’s rank among all $1 + I$ scores:

The Monte Carlo p-value and the decision

$$\hat{p} = \frac{1 + \#\{i \in \{1, \dots, I\} : Q_i \geq Q_{\text{obs}}\}}{1 + I}, \quad C = \mathbf{1}[V_{\text{obs}} = 1 \text{ and } \hat{p} < 0.05].$$

The bin clears when $C = 1$.

The 1 in the numerator counts the observed history as one of the draws, so $\hat{p}$ is never zero; with $I = 1,000$ the smallest possible value is $1/1,001$. Ties count against the bin. Unsupported observed bins are skipped, not tested. There is no multiple-testing correction in this rule; Stage 4 reports one beside it.

Unsupported synthetic runs push $\hat{p}$ down

On some synthetic histories this bin is unsupported and scores 0, which can never reach a positive Q_{obs} . Those runs stay in the count, which makes $\hat{p}$ smaller than if they were left out. The alternative counts only the $M = \sum_{i=1}^I V_i$ runs on which the bin was supported:

$$\hat{p}_{\text{supported}} = \frac{1 + \#\{i : V_i = 1, Q_i \geq Q_{\text{obs}}\}}{1 + M}.$$

The difference can decide the outcome. With $M = 100$ supported runs:

Supported runs $\geq Q_{\text{obs}}$	Current $\hat{p}$	$\hat{p}_{\text{supported}}$
4	5/1,001 ≈ 0.0050 clears	5/101 ≈ 0.0495 clears
5	6/1,001 ≈ 0.0060 clears	6/101 ≈ 0.0594 does not

Neither version is automatically right. The current rule compares against every synthetic history; the alternative compares only against histories on which this bin could be measured. The current rule is the implemented one, and each result records M so the gap can be checked. In both versions the null is fitted to the observed history, so $\hat{p}$ carries no exact guarantee.

Clearing is not discovery

At a 0.05 threshold, about one bin in twenty clears by chance when it has no effect. Among 1,000 tested bins with no effect at all, about 50 would clear. Clearing is an exploratory flag. Stage 4 therefore reports the count expected by chance and a q-value beside each bin (Section 7).

6.6 Worked example: applying the decision rule

To finish the running example, $Q_{\text{obs}} = 2.10$. Score 1,000 synthetic histories and suppose, for illustration, that 24 of them score at least 2.10. Then

$$\hat{p} = \frac{1 + 24}{1,001} \approx 0.025 < 0.05,$$

so the bin clears, and Stage 4 keeps its complete shift Surface. The count of 24 is assumed, not simulated: the observed Surface alone cannot tell how the synthetic histories will score. Other counts place the threshold:

Synthetic scores $\geq Q_{\text{obs}}$	$\hat{p}$	Clears?
24	25/1,001 ≈ 0.02498	yes
49	50/1,001 ≈ 0.04995	yes
50	51/1,001 ≈ 0.05095	no

The boundary sits between 49 and 50 synthetic scores at or above the observed one.

7 Stage 4: selection and presentation

Stage 4 adds no new test. It keeps the bins that passed, orders them by evidence, and says how much passing means when many bins are tested. From here on, the bins of all features are pooled: a *tested bin* b is one supported bin of one feature, with p-value $\hat{p}_b$.

7.1 Selection and rendering rules

Every bin that clears, and only those

$$\mathcal{K}_{\text{sel}} = \{b : C_b = 1\} = \{b : V_b = 1 \text{ and } \hat{p}_b < 0.05\},$$

where b ranges over all tested bins.

Each kept bin is drawn as its shift Surface heatmap beside the histogram of its 1,000 synthetic scores. Stage 3 draws the same figure for every tested bin.

There is no top- k cut, quota per feature, or minimum effect size. Kept bins are ranked by $\hat{p}$ alone; bins with equal $\hat{p}$ share a rank, and the score never breaks a tie. The order is evidence against chance, not size of effect.

How much would pass by chance

Let m be the number of tested bins. Two numbers put the selection in context; neither changes it.

Expected by chance

0.05 m : how many bins would clear if none had any effect and the test were exactly calibrated.

q-value

The Benjamini–Hochberg adjusted p-value: roughly, the share of bins at least this strong that would be false discoveries.

Sort the p-values of all m tested bins, not only the kept ones, as $\hat{p}_{(1)} \leq \dots \leq \hat{p}_{(m)}$. The bin at rank j gets

The q-value

$$q_{(j)} = \min_{l \geq j} \min\left(1, \frac{m \hat{p}_{(l)}}{l}\right).$$

The core is $\hat{p} \times m/\text{rank}$; the minimum over later ranks keeps q from falling as $\hat{p}$ rises, and bins with equal $\hat{p}$ share a q .

For example, with $m = 400$ tested bins, a bin ranked 10th with $\hat{p} = 0.01$ has $q \leq 0.01 \times 400/10 = 0.4$: of the ten strongest bins, about four would be expected by chance.

The q-value’s guarantee assumes independent or positively related tests. Here all features of one history share the same synthetic histories, and the simulator’s own calibration is not established (Section 6.3), so q-values give context, not a guarantee.

The number of synthetic histories limits q-values. With $I = 1,000$ the smallest possible $\hat{p}$ is $1/1,001$. If one of 400 tested bins reaches it and every other bin has $\hat{p} = 0.5$, that bin’s q-value is $400/1,001 \approx 0.40$. The strongest possible single result is still unremarkable at the family level.

7.2 Worked example: selecting the shift Surface

In the running example the bin is supported and its illustrative $\hat{p} = 25/1,001 < 0.05$ (Section 6.6), so it is kept, and its complete shift Surface is drawn:

$\delta \backslash \theta$	1	2	3	4	5	6	7
$Z_k = -2\%$	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10
-1%	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10
0%	0	0	0	0	0	0	0
$+1\%$	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10
$+2\%$	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10	+0.10

The score 2.10 summarized this Surface for the test; the output keeps all 35 cells.

8 Stage 5: combination of active conditions

Stage 5 adds no test and changes no selection. It asks what the kept conditions imply together for the last stored bar, and how far that combined estimate is from what history shows when those conditions held at once.

Stage 5 at a glance

$$\text{kept bins} \xrightarrow{\text{hold on the last bar}} \text{active} \xrightarrow{\text{one per family}} \text{used} \xrightarrow{\text{combine}} (P, J).$$

8.1 Active and used conditions

Let t_{last} be the last stored bar.

Active bin

A kept bin k of a feature whose value on the last bar falls in it: $x_{t_{\text{last}}}$ is finite and $k(t_{\text{last}}) = k$, under the feature's stored Stage 1 edges.

Used condition

The best-ranked active bin within its feature family, with ties broken by feature and then bin. Active bins from a history whose dates differ from the baseline's are not used. Every active bin that is not used is listed with the reason.

Write $\mathcal{C}$ for the used conditions and A_c for the dates on which condition $c \in \mathcal{C}$ held. Families group closely related features, such as one indicator at several periods, so the one-per-family rule removes their most obvious overlap. It does not remove overlap between different families.

8.2 Naive Bayes combination

At one barrier and horizon, let E be the event of a touch. Bayes' rule combines conditions through their odds:

$$\text{odds}(E \mid \bigcap_{c \in \mathcal{C}} A_c) = \text{odds}(E) \prod_{c \in \mathcal{C}} \frac{\text{odds}(E \mid A_c)}{\text{odds}(E)}.$$

Each condition multiplies the baseline odds by how much it changes the odds on its own. This is exact only when the conditions are independent once the outcome is known.

Rates are smoothed first, as if each group had one extra touch and one extra miss, so that a rate of 0 or 1 still has finite odds:

$$\tilde{p} = \frac{\text{touches} + 1}{\text{dates} + 2}.$$

Taking logarithms turns the product into a sum. With $\text{logit } p = \log(p/(1-p))$:

Naive Bayes: each condition adds its log odds ratio

$$\text{logit } P(\delta, \theta) = \text{logit } \tilde{B}(\delta, \theta) + \sum_{c \in \mathcal{C}} (\text{logit } \tilde{Y}_c(\delta, \theta) - \text{logit } \tilde{B}(\delta, \theta)),$$

so $P \in \mathcal{S}$, built cell by cell.

Its support is the weakest contributor at each horizon,

$$n^P(\theta) = \min\left(N(\theta), \min_{c \in \mathcal{C}} N_c(\theta)\right),$$

and a cell of P is blank when $n^P(\theta) < n_{\min}$. With no used conditions, $P = \tilde{B}$. Conditions that measure the same thing count the same evidence more than once, which pushes P further from the baseline than the evidence supports.

8.3 The historical joint rate

The direct estimate needs no independence assumption:

The historical joint rate

$$J = \Pi \left[\bigcap_{c \in \mathcal{C}} A_c \right] \in \mathcal{S},$$

the unsmoothed touch rate over the dates on which *every* used condition held.

It is the same operator Π that produced Y_k and B , so it is blank below $n_{\min}$ such dates. With no used conditions, the intersection is all of $\mathcal{T}$ and $J = B$.

P and J trade one weakness for another:

Naive Bayes Surface P

Uses each condition's full sample, but assumes the conditions are independent given the outcome.

Historical joint Surface J

Assumes nothing about dependence, but its sample shrinks quickly: two unrelated decile conditions hold together on about one date in a hundred.

The gap $P - J$, summarized by its average and its largest value, shows how far the independence assumption moves the estimate, read against how few dates J rests on.

Both are in-sample. Every used bin was kept because its score was extreme, so its measured shift is likely larger than its true effect, and neither P nor J corrects for that.

8.4 Nesting of touch events

A farther barrier can only be reached by passing a nearer one on the same side, and a touch within θ bars is also a touch within any longer horizon. Any Surface F of touch probabilities must therefore satisfy

Nesting inequalities

$$\begin{aligned} F(\delta', \theta) &\leq F(\delta, \theta) && \text{for } 0 \leq \delta < \delta' \text{ or } \delta' < \delta < 0, \\ F(\delta, \theta) &\leq F(\delta, \theta') && \text{for } \theta < \theta'. \end{aligned}$$

The measured Surface of Section 4.4 respects this; a combined Surface need not:

Barrier	Nested (measured Y_k)	Violating (illustrative)
0%	0.90	0.90
+1%	0.45	0.45
+2%	0.35	0.48 ↑

In the right column the +2% barrier is more likely to be touched than the nearer +1% barrier, which is impossible for real touch probabilities.

Stage 5 counts the adjacent pairs of cells of P that break these inequalities: neighbouring barriers within $\delta < 0$, neighbouring barriers within $\delta \geq 0$, and neighbouring horizons, ignoring blank cells. It reports the count without correcting it. Adding log odds ratios cell by cell does not have to preserve the order. Even measured rates can break the horizon order slightly, because longer horizons leave out the most recent dates.

8.5 Strongest cell of a used condition

For each used condition, the report also points to one cell worth looking at: the largest shift inside a run of at least two neighbouring barrier rows that all move the same way by at least 10 percentage points, at a horizon where the bin has at least 50 dates (the NASDAQ defaults). It is a reading aid for the shift Surface; nothing is tested.

8.6 Worked example: combining two conditions

At one cell, suppose the baseline has 250 touches among 1,000 dates, condition 1 has 40 among 100, and condition 2 has 35 among 100, from different families. Smoothing and log-odds give

	Smoothed rate $\tilde{p}$	logit $\tilde{p}$	Log odds ratio
Baseline	251/1,002 ≈ 0.2505	-1.0960	
Condition 1	41/102 ≈ 0.4020	-0.3973	+0.6987
Condition 2	36/102 ≈ 0.3529	-0.6061	+0.4898

so that

$$\text{logit } P = -1.0960 + 0.6987 + 0.4898 = 0.0925, \quad P = \frac{1}{1 + e^{-0.0925}} \approx 0.523.$$

The smallest group behind it has 100 dates. Each condition alone raises the touch rate from about 25% to 35–40%; together, under independence, they raise it to about 52%.

Now suppose both conditions held on 40 dates and the barrier was touched after 15 of them. Then $J = 15/40 = 0.375$, enough dates to report, and the gap is $P - J \approx +0.148$. Side by side:

Estimate	Touch probability	Dates behind it
Baseline $\tilde{B}$	0.2505	1,000
Naive Bayes P	0.523	100
Historical joint J	0.375	40

History with both conditions looks much like history with either one, as it would if the two conditions largely measured the same thing. The joint rate rests on only 40 dates, so the gap is itself uncertain.

9 Short reference

Conventions and defaults behind the explanations above.

Item	Convention
Bin edges	Linearly interpolated quantiles of the finite feature values; repeated edges removed; a value on an edge goes to the lower bin.
Minimum dates	30 per cell for rates, shifts, scores, and P and J ; 50 for the strongest-cell line.
Mirror pairs	A barrier counts only with exactly one mirror of opposite sign on the grid.
Synthetic histories	$I = 1,000$ per distinct market history, seed 20260907, shared by every feature of that history. CPU and CUDA generators give different, equally valid ensembles for the same seed.
Null fit	Rows with a non-finite component are dropped; the first bar uses its own close as the previous close; the covariance is the $n - 1$ sample covariance plus diagonal jitter $10^{-12} \max(1, \max_j \Sigma_{jj})$ before the Cholesky factorization.
Unsupported synthetic runs	Score 0 and stay in the p-value count.
Clearing	Supported and raw $\hat{p} < 0.05$; no multiple-testing correction in the rule.
Default NASDAQ grid	Barriers -20% to $+20\%$ in 1% steps (41), horizons 1–30 bars, 10 bins.
Stored precision	Stored probabilities and shifts are float32; Stage 3 remeasures every score in float64 from the stored prices.

Stored arrays use these axes:

Quantity	Array axes, in order
OHLCV histories	history, bar, OHLCV component
Touches at one horizon	history, bar, barrier
Conditional probabilities and hit counts	history, barrier, bin, horizon
Baseline probabilities	history, barrier, horizon
Shift surfaces	history, barrier, bin, horizon
Bin date counts	history, bin, horizon
Bin scores and support flags	history, bin